

Notations

Reference for the notations used throughout **Linear Algebra for Machine Learning and Data Science** (DeepLearning.AI, Course 1). Do not worry if some are unfamiliar yet; each is covered as the course progresses.

Notation	Meaning
A, B, C	Capital letters represent matrices
u, v, w	Lowercase letters represent vectors
A of size $m \times n$ or $(m \times n)$	Matrix A has m rows and n columns
A^T	The transpose of matrix A
v^T	The transpose of vector v
A^{-1}	The inverse of matrix A
$\det(A)$	The determinant of matrix A
AB	Matrix multiplication of matrices A and B
$u \cdot v; \langle u, v \rangle$	Dot product of vectors u and v
$\mathbb{R}$	The set of real numbers, e.g. $0, -0.642, 2, 3.456$
$\mathbb{R}^2$	The set of two-dimensional vectors, e.g. $v = [1 \ 3]^T$
$\mathbb{R}^n$	The set of n -dimensional vectors
$v \in \mathbb{R}^2$	Vector v is an element of $\mathbb{R}^2$
$ v _1$	L1-norm of a vector
$ v _2; v ; \ v\ $	L2-norm of a vector
$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3; T(v) = w$	Transformation T of a vector $v \in \mathbb{R}^2$ into the vector $w \in \mathbb{R}^3$